Wen Qi

PhD Candidate in Geography, University of Maryland | Research Trainee, Harvard Medical School

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Research Interests

Precision Medicine • Digital food platform and food access • Spatial epidemiology and optimization

Education

Expected 2027 **PhD, Geographical Sciences**, University of Maryland, College Park.
Dissertation: *The Health-Environment Tradeoff in U.S. Diets: Inequity Across Generations And Space*.
Committee: Giovanni Baiocchi (chair), Laixiang Sun, Taylor Oshan, Pan He (University of Hong Kong), Edmond Shenassa (UMD School of Public Health).

2023 **MS, Finance**, University of Maryland, College Park.
Quantitative training in econometrics, financial modelling, and optimization.

2020 **BS, Finance**, Hunan University

Appointments

2026–present **Research Trainee**, Channing Division of Network Medicine, Brigham and Women's Hospital and Harvard Medical School, Boston, MA.
Mentor: Giulia Menichetti.

2022 **Research Assistant**, UN Trade and Development (UNCTAD), Switzerland.

Publications

Qi W = *candidate.*

Peer-Reviewed Journal Articles

1. **Qi W**, Baiocchi G. When healthier eating emits more: socioeconomic divergence in US diet sustainability. *Journal of Cleaner Production*. 2026;575:149143. doi:10.1016/j.jclepro.2026.149143

Manuscripts Under Review

1. **Qi W**, Sun L, Baiocchi G, Shenassa E. Younger birth cohorts have poorer diets but greater health gains from plant-rich dietary shifts. *Under review*.

Manuscripts in Preparation

1. **Qi W**, Baiocchi G. Food complexity and Ultra-processed food prevalence across the world.
2. **Qi W**, Yao Li, Baiocchi G, Sun L, Hsia RY, Shen YC, Shenassa E. Misalignment between Acute Myocardial Infarction and access to certified emergency care in the United States: a national spatial analysis and network optimization study.
3. Sebek M, **Qi W**, Jain A, Nicholi F, Ispirova G, Menichetti G. Bert-Finetuned Model updates about FPro and GroceryDB extension.

4. **Qi W**, Baiocchi G. Nutrition-Blind Shelves: Auditing US Online Baby Food against the WHO Nutrient and Promotion Profile Model.

Reports and Other Publications

1. UNCTAD. *The Least Developed Countries Report 2022: The Low-Carbon Transition and Its Daunting Implications for Structural Transformation*. Switzerland; 2022. [Report]

Grants, Fellowships & Awards

Research Grants

2026–2027 **MPower Early Scholars Investment Fund**, Maryland State ($19,167).
Spatial mismatch between stroke care access and population needs.

2025–2027 **Dean's Research Initiative Grant**, University of Maryland ($3,800).
Structural determinants of cardiovascular and cardiometabolic disparities: food environments and neighborhood food access. Supports data acquisition.

Fellowships and Awards

2025 Human Dimensions of Global Change Summer Research Fellowship, Department of Geographical Sciences, UMD ($5,000).

2024 Jacob K. Goldhaber Travel Award, Graduate School, UMD ($800).

2024 Graduate Travel Fund, Department of Geographical Sciences, UMD ($500).

2023 Dean's Fellowship, Graduate School, UMD ($19,320).

Technical Skills

Machine Learning: XGBoost, random forest, fine-tuned BERT; classification of 100,000+ retail products.
Research Methods: Model evaluation, causal inference (DiD, matching, panel data), survey-weighted regression, age–period–cohort models, and time-series analysis.
Programming & Tools: Python, R, SQL, Stata, SAS, MATLAB, ArcGIS Pro, QGIS, and HPC clusters.
Optimization: Linear and quadratic programming, network analysis, location–allocation models, and cost-effectiveness modelling.
Data: Healthcare and Utility Project, NHANES, ACS, FNDDS, Open Food Facts, and scraped retail data.

Personal Project

SmartCook 2026
Bilingual iOS meal-planning and grocery app; sole developer; App Store–ready

- **Constraint-based planner:** Turns dietary goals, allergies, household size, budget, and pantry inventory into weekly gram-level meal plans and a consolidated grocery list. Expiry-aware "fridge-first" planning helps reduce household food waste.
- **Safety-first nutrition engine:** Uses Mifflin–St Jeor energy estimates with MET-based activity adjustment. Property-based tests verify hard allergen exclusions.
- **Native iOS integration and release:** Implemented MapKit store matching and local notifications; debugged WebKit/Capacitor integration; prepared the privacy label, health positioning, and TestFlight release.

Vision-Guided Pick-and-Place with a UR3e Arm 2026

ROS, OpenCV, Python

- **Camera-to-robot pipeline:** Calibrated the camera, detected AR-tagged blocks with OpenCV, computed centroids, and mapped pixel coordinates to the robot's world frame using a perspective transform.
- **Robot motion:** Implemented forward and inverse kinematics for the UR3e and ROS nodes to move three detected blocks to target poses.

Presentations

Conference Presentations

2026 *Research Day*, School of Public Health, UMD.

2026 *Graduate Research Appreciation Day*, UMD.

2024 *AGU Fall Meeting*, Washington, DC.

2024 *American Association of Geographers Annual Meeting*, Honolulu, HI.

Teaching

Instructor

GEOG 202: Introduction to Human Geography
Summer 2025, Winter 2025, Summer 2026, Winter 2026 (scheduled)

Teaching Assistant

GEOG 306: Introduction to Quantitative Methods for the Geographical Environmental Sciences
GEOG 170: Mapping the Digital World
GEOG 140: Natural Disasters: Earthquakes, Floods, and Fires

Service & Memberships

2026–2027 Student Board Member, International Society for Industrial Ecology (ISIE).

Member: American Association of Geographers (AAG); American Geophysical Union (AGU); International Society for Industrial Ecology (ISIE)

References

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